

# The Gate and the First Flow: Program $\omega$ , Stage C's Opening Move

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**Revision 1.4 — closed on all four discriminators and the branch-sector correspondence (T-2); open only on the interpretation session. Companion to the frozen  $\omega$ -plan (Stage B closed on the two-sign question; this note covers the Gaussianity gate, the arithmetic fork it surfaced, the first flow integration on two anchors, the frozen-sector class-theorem, discriminator 2's own band grid, and the branch-sector correspondence's resolution — all now two-implemented). Every claim below is two-implemented unless explicitly marked otherwise; the error ledger is part of the record because the catches, across this note's five revisions, are the substance of what was learned.**

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## 1. Charter

Stage B closed with  $m(u)$ 's asymptotic signs undetermined-without-dynamics — the missing datum handed forward to Stage C. This note covers Stage C's own opening: constructing the Gaussianity gate the frozen plan's P3 amendment specifies, the extended arithmetic fork that construction surfaced, and the first flow integration once the fork closed. It ends without a verdict, by design — the two anchors integrated here produced genuinely different, only partially interpreted behavior, and that openness is the honest state of the record, not a placeholder for a conclusion not yet written.

## 2. The Gaussianity gate: construction and its two-implementation

**[FACT, two-implemented]** The candidate dissipative dynamics is built from linear jump operators on the anchor's own real-linear operator  $L_A$ , expressed in a mode basis built from  $J_0 = -L_{e_4} - \mathbb{S}$ 's own invariant complex structure, fixed by the mediator direction, distinct from any wall-sector or gauge-module object also carrying the name " $J_0$ " elsewhere in the corpus. (This distinction is stated explicitly here because blurring it was the exact species that caused this arc's own stale-coordinate bug, §3 below; a note about that bug should not commit it.) In quadratures  $r = (x_1, p_1, \dots)$ , each jump  $L_j = c_j^T r$ , with  $M = \sum_j c_j c_j^\dagger$ , the covariance obeys the closed-form Lyapunov equation

$$\dot{\sigma} = A\sigma + \sigma A^T + D, \quad A = -\Omega \operatorname{Im}(M), \quad D = \Omega \operatorname{Re}(M) \Omega^T,$$

$\sigma$  centered and symmetrized,  $\Omega$  the standard symplectic form. This formula was independently derived and cross-verified against direct Fock-space simu-

lation (finite truncation, exact master-equation dissipator), converging to machine precision as truncation grows — the formula itself was never in dispute once both sides tested it against ground truth rather than each other.

### 3. The arithmetic fork, and what it actually was

**[LEDGER, the arc’s own record]** What followed was not a physics disagreement but a five-layer convention stack, each layer catching a genuine error before the next could hide behind it.

**Layer 1 — stale coordinates.** The anchor  $L_A$  built fresh in this session’s own basis was combined with a saved  $J_0$  file from an earlier part of this long session, built on a silently different basis realization. Both objects passed every intrinsic check individually ( $J_0^2 = -\text{Id}$  exact;  $L_A$ ’s tunnel spectrum exact) — alignment between them was never tested until a full rebuild on consistent inputs was forced. Root cause named precisely: **saved-object coordinate provenance** — a file can carry validity without carrying the coordinate frame it was validated in, and two individually-correct objects from different moments of one session are not automatically combinable. Resolution: full rebuild from one `cd16` call, no saved files. Confirmed split:  $\|C\|^2 = 4$ ,  $\|D\|^2 = 12$ , matching the independently-derived basis-free invariant ( $\|L_c\|^2/2$ ,  $\|L_a\|^2/2$  for the  $J_0$ -complex-linear/antilinear parts of  $L_A$ ) exactly.

**Layer 2 — the definitional fork.** Two constructions for “the anchor’s dynamics under the mirror orientation” were each built and verified exact: (a) the same quadrature machine fed the mirror’s conjugated inputs, giving  $A_2 = A_1$  (orientation-blind) and  $[P, A_1] = 0$ ; (b) the literal adjoint jump-set  $\{L_j^\dagger\}$ , giving  $A_2 = -A_1$  (arrow-duality, loss $\leftrightarrow$ gain). Both are exact theorems about two different, well-defined objects — not competing answers to one question. **[THM, resolved as a duality, not a choice]** The two compose: orientation-flip followed by the construction equals time-reversal of the construction. Neither reading is privileged; a future document naming “the mirror candidate” states locally which half it means.

**Layer 3 — a mislabeled scalar.** A proposed “kernel diffusion =  $N(A)$ ” identity was traced to its source and found to be a Frobenius norm of a  $4 \times 4$  block, reported as a per-direction rate. Rerun with the value computed two independent ways (via the assembled  $D$  matrix, and directly as  $\sum_j |c_j \cdot k|^2$ ) gives  $k^T D k = 1.0$  per kernel direction, both routes agreeing to machine precision. Bonus structural fact, confirmed:  $\text{Re}(M)$  annihilates the kernel directly — the jump vectors are exactly orthogonal to it — so the diffusion the kernel experiences arrives entirely through the  $\Omega$ -dressing, from its symplectic partner directions, not from any direct coupling.

**Species logged, this arc, both sides:** *saved-object coordinate provenance* (mine); *output-attribution blur*, *rationalized disconfirmation*, *theorem-scope in-*

flation, quantity-type mislabeling (F's, each independently self-diagnosed and ledgered in turn). The countermeasure common to all: a cheap, mechanical check (print the actual matrices consumed, not re-derived ones) run *before* trusting an internally-consistent-looking result, not after a contradiction forces it.

#### 4. Six results, closed

Confirmed two-implemented, on fully consistent inputs, no open threads:

1. **Split:**  $\|C\|^2 = 4$ ,  $\|D\|^2 = 12$ ,  $\text{tr}(A) = 8$ .
2. **Drift spectrum:**  $\{0^8, +1^8\}$  for the classic anchor  $e_1 + e_{10}$ .
3. **D-equality:**  $D_2 = D_1$  under either reading of the orientation flip.
4. **Kernel drift-freeze:** exact to  $10^{-16}$ , refined to  $\text{Re}(M)$ -annihilation.
5. **The 84-diagonal survey:** exactly three classes, 12/60/12, trace  $\in \{+16, +8, -8\}$ .
6. **The strut cross-tabulation:** the mediator kite is uniquely class-pure (all 12 diagonals in the middle class); every other kite splits identically  $2+8+2$  across the three classes, verified via the seam note's own assessor-product strut rule.

**[THM] The survey is no longer surveyed — derived.** Each basis unit carries an individual, cross-talk-free ( $\|L_c\|^2, \|L_a\|^2$ ) contribution —  $e_4$ :  $(16, 0)$ , purely complex-linear ( $L_{e_4} = -J_0$  trivially);  $e_{12}$ :  $(0, 16)$ , purely antilinear ( $\{L_{e_{12}}, J_0\} = 0$  exactly); generic  $\text{Im}(\mathbb{O})$ :  $(0, 16)$ , via the Formal Beginning's own  $\text{Cl}(8)$  structure ( $\{L_{e_i}, L_{e_j}\} = -2\delta_{ij}$  for  $i, j \leq 8$ , so every imaginary octonion unit anticommutes with  $L_{e_4}$ ); generic  $\mathbb{O}e_8$ :  $(8, 8)$ . Every diagonal anchor's class is the halved sum of its two ingredients — reproducing the 12/60/12 census identically, with the extreme-pair identification and the fourth combination's absence ( $e_4 \cdot e_4 = -e_0$ , not an imaginary unit, so the  $e_4 + e_{12}$  plane is never an assessor at all) both falling out as corollaries rather than separate facts.

**A seventh, discovered by the flow rather than the gate:** the linear-growth branch's slope in the contracting anchor's own  $\chi_{\text{kin}}(t)$  trajectory equals the gate's certified kernel diffusion constant exactly — 1 per direction per unit time — two independently-certified numbers, one from the gate's static analysis and one from the flow's dynamics, agreeing across the gate/flow boundary with no free parameter connecting them.

#### 5. The first flow: two anchors, three discriminators, two-implemented

**[FACT, two-implemented]** Two anchors integrated from a shared quasi-free initial state ( $\nu = 1.5$ ): the classic expanding-class anchor ( $e_1 + e_{10}$ ,  $\text{tr } A = +8$ )

and a contracting-class anchor ( $e_4 + e_9$ ,  $\text{tr } A = -8$ , drift spectrum  $\{-2^4, 0^{12}\}$  — a genuinely different structure from a simple sign-flip of the expanding case). An independent closed-form integration (a different ODE route — the exact matrix-exponential solution rather than numerical quadrature — and a different symplectic-spectrum implementation) reproduced every landmark digit-for-digit: the expanding branches  $\{1.534, 6.078\} \rightarrow \{2.373, 1139.067\}$ ; the contracting three-branch structure with multiplicities  $2/2/4$  — the vacuum-landing at exactly 0.5, the branch frozen at exactly 1.5 throughout, and the linear branch as  $\nu = 1.5 + t$  exactly; the family-residual growth rates, consistent up to metric convention.

**Discriminator 1 (deepens vs. wanders), stated as result.** The expanding anchor's  $\chi_{\text{kin}}$  **deepens**: both branches grow monotonically, every successive state comparable and strictly deeper in Becoming's own partial order — genuine, coordinated, spectrum-valued depth-motion, off-spine, without ever reaching stationarity. The contracting anchor **wanders**: its three branches move incomparably — one decaying, one static, one growing — simultaneously in all three senses, which is wandering by the same criterion applied honestly. The anchor named for stability (by its drift matrix) is the one that wanders; the anchor that deepens cleanly never rests. Given the corpus's own direction convention (the vacuum end as the unattained past), the expanding anchor's behavior is deepening without completion — the no-completion doctrine, asserted as state-theory since Foundation's own boundary facts, realized here by a dynamics nobody tuned to produce it.

**Discriminator 3 (family-tracking).** Neither anchor stays on the quasi-free line; both depart it immediately upon integration, at rates differing by roughly two orders of magnitude — the expanding anchor's departure explodes, the contracting anchor's is comparatively gentle.

## 6. The frozen sector: identified, mechanism included

**[FACT, two-implemented]** The contracting anchor's flow carries a genuinely silent sector — four directions with both zero drift and zero diffusion, frozen at their initial value throughout the integration. A neutral scan against every natural candidate subspace (no favored hypothesis) identifies these exactly: **the frozen directions coincide precisely with  $L_A$ 's own kernel** — the seam note's tunnel — with maximum overlap (2.0000 against a 4-dimensional space) and exact zero against every other candidate tested (the scalar span, the anchor plane, the strut fiber). The split against the  $\tau$ -halves is exactly equal, matching the tunnel lemma's own “perfectly  $\tau$ -balanced” description precisely. The expanding anchor carries no such sector at all — every one of its zero-drift directions carries identical, nonzero diffusion; the frozen phenomenon is not generic to the construction.

**[LEMMA, two-implemented] The mechanism.** For  $k \in \ker L_A$ ,  $L_c k =$

$-L_a k$  always (since  $L_c + L_a = L_A$  vanishes on the kernel) — but the kernel is  $J_0$ -invariant if and only if the stronger, joint condition holds:  $L_c k = L_a k = 0$  *separately*. Verified across all 84 canonical diagonals: at the pure-antilinear class ( $C^2 = 0$ ), this is automatic; at the genuinely mixed extreme class ( $C^2 = 12, D^2 = 4$  — *not* pure, both parts nonzero), the joint annihilation is a real, nontrivial fact, confirmed exactly ( $\|L_c \cdot \ker\| = \|L_a \cdot \ker\| = 0$  on all 12 members); at the generic middle class, it fails on all 60. **The seam note’s own tunnel lemma — kernel commutes with neither  $\tau$  nor  $L_{e_4}$  — is generic, not universal:** the exceptional set is exactly these 24 extreme-class diagonals, both classes, not one.

**[FACT, two-implemented] The freeze is sign-blind by the shape of its own formula, not by a check that could have failed.**  $J_0$  enters  $L_c, L_a$  only sandwiched —  $(-J_0)X(-J_0) = J_0 X J_0$  identically for any  $X$  — so the split, the three classes, and the 24-anchor criterion could never have distinguished  $J_0$  from  $-J_0$  at any stage, before any anchor is even chosen. (An earlier framing of this as “the sanctuary’s door is keyed to the mirror bit and happens to pass” is retired: the check that phrasing describes is a tautology for any subspace, structurally incapable of failing, and tests nothing anchor-specific. The correct, stronger statement is that nothing here was ever at risk.)

**[THM, two-implemented] The class-prediction is now a class-theorem.** A full sweep confirms both drift-freeze and diffusion-freeze on all 24 extreme-class anchors, not merely the one first tested (max residuals  $5.7 \times 10^{-15}$  and  $1.2 \times 10^{-16}$  across the full set). Separately,  $\text{Re}(M)$ ’s annihilation of the kernel is confirmed universal — true on all 84 diagonals, including the 60 generic ones — following directly from the jump vectors being built from  $L_A$ ’s own rows, which the kernel kills by definition; the diffusion-freeze’s restriction to exactly the 24 extreme-class anchors comes entirely from whether the kernel’s symplectic partner ( $\Omega^T k$ ) itself stays inside a region  $\text{Re}(M)$  annihilates — which is exactly the  $J_0$ -invariance condition above.

**[FACT, two-implemented] Question (iv), answered honestly negative.** The scalar directions ( $e_0, e_8$ ) were tested directly against each anchor’s own drift/diffusion structure, no hypothesis presumed: both carry nonzero drift and nonzero diffusion in every anchor tested — confirmed on the two anchors of §5 at the time, and confirmed again, later, on the  $(0, 16)$  anchor of §6 once it existed — a representative of all three classes now checked, not merely two. **The dissipation does not conserve the scalar channel** — it engages the measurement channel exactly as it engages any generic direction, in every class tested. The kernel avoids the scalars statically (the scalar-avoiding kernel theorem stands, untouched); the dynamics does not.

## 7. Discriminator 2: the drift matrix's own invariant decomposition

**[FACT, two-implemented] The band grid, measured.** Testing directly whether  $A$  maps each  $T_A$ -band into itself (basis-independent, required since  $A_{\text{exp}}$  is non-normal —  $\|AA^T - A^TA\| = 4\sqrt{2}$ , versus  $\sim 0$  for  $A_{\text{con}}$ ): the **contracting anchor is fully block-diagonal** across all three bands (kernel frozen; band 2  $\rightarrow$  band 2 exactly; band 4  $\rightarrow$  band 4 exactly, no cross-terms). The **expanding anchor's band 4 does not stay within itself** — it maps into a combination of band 4 and the kernel (equal split, zero into band 2), while band 2 remains cleanly self-contained in both anchors.

**[FACT, two-implemented] The sector spectra, measured.** For the expanding anchor, both invariant sectors — band 2, and the coupled kernel+band-4 sector — carry the identical mixed spectrum  $\{1^4, 0^4\}$  (trace 4 each, summing to the full trace 8). The coupled sector is confirmed **not nilpotent**:  $\|A_{\text{coupled}}^n\| = 2.828$ , unchanged from  $n = 1$  through  $n = 5$ .

**[RESOLVED, T-2] The correspondence between the drift's invariant sectors and the flow's spectral branches, closed by two mechanisms.** Both earlier proposed stories died on their own numbers rather than each other's: a nilpotent-tower reading (predicting polynomial slow-branch growth) failed once the coupled sector's actual, non-vanishing spectrum was checked; a same-qualitative-growth reading (predicting  $e^6 \approx 403$ -fold growth by  $t = 3$  for any +1-eigenvalue-bearing sector) failed against the measured slow branch (2.373 at  $t = 3$ , a factor of 170 short). The actual mechanism, found via the kernel-invariance criterion's own symplectic reading (G3): a band is  $\Omega$ -nondegenerate iff  $J_0$ -invariant, Lagrangian iff  $J_0$ -transverse. For this (generic) anchor, all three bands are  $J_0$ -transverse, and the drift/diffusion-invariant decomposition (band 2, the coupled kernel+band-4 sector) is a genuine Lagrangian pair — the observed branches are the geometric mean  $\nu(t)^2 = \text{eig}(\sigma_2(t) \cdot B \cdot \sigma_c(t) \cdot B^T)$  across the seam between them, confirmed to four decimals against the measured  $\{2.373, 1139.067\}$ . For the extreme-class anchors, by contrast, all three bands are individually  $J_0$ -invariant and  $\Omega$ -nondegenerate — each a self-contained phase space, its own restricted symplectic spectrum reproducing the anchor's own branches exactly (contracting anchor: kernel  $\rightarrow$  1.5 frozen, the 2-band  $\rightarrow$  4.5, the 4-band  $\rightarrow$  0.5, all confirmed). See the T-2 note for the full derivation, both mechanisms, and the honest scope (one anchor per class, not an 84-anchor sweep).

## 8. Open items, carried forward

- The contracting anchor's three-way split (decay / frozen / linear growth) is now fully explained by the same mechanism as the expanding anchor's own correspondence — see the updated §7 above and the T-2 note.

- The interpretation session itself: with all four discriminators' data assembled, whether either anchor's behavior — or some structural feature short of either — constitutes "settling" in the corpus's own sense has not yet been asked.
- Whether either anchor's behavior should inform REG-2's successor question (the anchor-supplier problem) remains entirely untouched by this note.
- Two downstream touches flagged, not yet made: the theorem inventory gains an entry for the kernel-invariance criterion (§6); the seam note's tunnel lemma gains a one-line generic-not-universal scope refinement, at next opening.

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